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Longitudinal Remote 6-Minute Walk Distance and Cardiac Effort Using Wearable Sensors in Pulmonary Hypertension

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ABSTRACT

Remote 6-min walk testing (6MWT) has been shown to be safe in pulmonary hypertension, but limited data exist on its longitudinal durability and relationship to supervised in-clinic testing. Cardiac Effort (CE), defined as total heartbeats divided by walk distance, provides physiologic context beyond distance alone. We evaluated the longitudinal performance of a home-based 6MWT and CE performed on a modified indoor walking space using a wearable chest sensor and compared results with in-clinic testing over 8 weeks. This single-center prospective study was conducted with institutional review board approval. Participants with PH performed supervised in-clinic and unsupervised home 6MWTs at baseline and Week 8. A chest-based wearable sensor recorded acceleration and heart rate data to estimate walk distance and calculate CE. Mixed-effects repeated-measures models were used to compare sensor-estimated and directly observed walk distance and CE in the clinic and at home over time. Agreement was assessed using Bland–Altman analyses, intraclass correlation coefficients (ICC), and Spearman correlations. Twenty-nine participants were enrolled, and no adverse events occurred. Sensor-estimated 6MWT distance and CE demonstrated minimal bias and excellent agreement with directly observed in-clinic measurements (6MWD bias 0.04%, ICC 0.995; CE bias −0.04%, ICC 0.995). Home-based 6MWD was approximately 40 m lower than supervised clinic testing. Mixed-effects models showed similar longitudinal changes between home and clinic testing. Changes in home 6MWD and CE correlated with corresponding in-clinic changes (6MWD: $r = 0.40$, $p = 0.045$; CE: $r = 0.42$, $p = 0.03$). Home-based 6MWT and CE obtained using a wearable chest sensor provide a safe, feasible method for remote functional assessment in PH that tracks with in-clinic testing. Frequent home assessment may facilitate earlier detection of clinical change, support therapeutic titration, and expand access to standardized physiologic monitoring.

The 6-min walk test (6MWT) is a simple, widely used assessment of functional capacity [1]. In some patients with pulmonary arterial hypertension (PAH), the physiologic stress achieved during the 6MWT is comparable to maximal cardiopulmonary exercise testing [2]. The test requires only minimal equipment, an unobstructed walking space, and personnel to measure laps and calculate the 6-min walk distance (6MWD). Because 6MWD is deeply integrated into PAH management and risk assessment [3], it remains a cornerstone clinical endpoint. However, the frequency with which 6MWT can be performed is limited, particularly for patients who live far from specialized PAH centers. Cardiac effort, the number of heartbeats expended during the 6MWT divided by the distance walked, was developed to provide deeper insight into the physiologic toll required

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to achieve a given 6MWD [4–8]. Interest in cardiac effort has grown, and it has already been incorporated as an exploratory endpoint in a PAH clinical trial [9], in pulmonary hypertension with interstitial lung disease (NCT06129240), and studies evaluating recovery after pulmonary embolism [10] and PAH rehabilitation (NCT06477640).

The COVID-19 pandemic disrupted routine clinical care for patients with PAH, and interpreting changes in 6MWD became more challenging due to masking requirements [11] and the rapid expansion of telemedicine, which limited opportunities for directly observed functional testing. As a result, several groups evaluated the feasibility of performing the 6MWT remotely [7, 12, 13]. Across these studies, home-based 6MWT was consistently shown to be safe for patients with PAH. The first report demonstrated that home and clinic 6MWD were comparable when the test was performed in a 90-foot walking space under semi-supervised conditions [12]. A second study showed that unsupervised 6MWT conducted in a shorter indoor space (~40 ft) yielded lower home 6MWD when estimated by wearable sensors; however, cardiac effort provided similar measurements between home and clinic settings [14]. A third study used a smartphone-based application to perform tele-6MWT in a mixed pulmonary hypertension cohort and found that the app overestimated 6MWD by approximately 25 m compared with directly observed testing [13]. Together, these findings highlight both the feasibility and limitations of current approaches to remote 6MWT and underscore the need for accurate, physiologically grounded home-based assessments.

In this study, we evaluated supervised clinic and unsupervised modified home 6MWT and cardiac effort in a mixed cohort of patients with PAH and chronic thromboembolic pulmonary hypertension at baseline and again 8 weeks later. We hypothesized that (1) cardiac effort, by accounting for heart rate changes, would provide more comparable measurements between clinic and home settings than 6MWD, and (2) changes in home-based cardiac effort would correlate more closely with changes observed in the clinic at the 8-week follow-up than would changes in home 6MWD.

1 | Methods

This was a prospective observational study conducted at the University of Rochester Medical Center. Institutional Review Board approval was obtained. All participants provided informed consent before any study activities were performed. Participants with treated PAH or chronic thromboembolic pulmonary hypertension were eligible to participate. Recruitment occurred between September 2022–March 2023 as they were seen during routine clinic visits. Participants needed to be functional class I-III, ambulatory, and have an unobstructed walking space of at least 30 ft at home.

Baseline testing occurred during a separate visit and included demographics, clinical characteristics, quality of life questionnaire (emPHasis-10 [15]), NT-pro BNP, and Reveal lite 2 score [16]. The BioStamp nPoint (Medidata), a medical-grade wearable patch that has been previously validated to measure accelerometry and electrocardiography [17] and used in this pilot study [7], was placed on the left upper chest and thigh with disposable adhesive prior to the 6MWT. Accelerometry and electrocardiography data were recorded at 31.25 and 250 Hz, respectively. Concurrently, we utilized the VitalPatch (VitalConnect), a disposable wearable patch also measuring accelerometry (50 Hz) and electrocardiography (125 Hz), validated for remote vitals data collection [18, 19]. Both devices are FDA-cleared for medical use. Due to supply chain issues, participants wore BioStamp nPoint or VitalPatch. Participants used the same device during both monitoring periods. Data from both devices was collected and transmitted to a secure cloud for analysis. The goal was not to compare sensors as they were used based on availability, and the study focus was to be device agnostic. Participants rested for 3 min before the clinic 6MWT. At each visit (Day 0), participants completed directly supervised unmasked 6MWT [1]. On Days 1 and 2, a home 6MWT on a modified walking space was performed. The 2nd home 6MWT was used for analysis to account for the learning effect or technical issues. The course length was recorded by the participant using a 100 ft tape measure provided. Safety assessment was performed to evaluate for syncope, falls, or other safety concerns related to home 6MWT or devices. Eight weeks (+/− 7 days) later, participants would complete the same evaluation as baseline. Due to COVID scheduling restrictions, participants could only follow up in the allotted window. If they could not make the in-window testing, the Week 8 assessment was missed.

6MWD in clinic was calculated by (1) directly observed distance ($6MWD_D$) and (2) estimating distance based on lap counting with accelerometry (as previously described) [7]. In the home setting, 6MWD was estimated using lap counting only. Cardiac effort (CE) was calculated as previously described [5, 7] using both 6MWD methods. The electrocardiography heart rate data was analyzed using LabChart 8 (AdInstruments).

Categorical variables were reported as counts and percent. Continuous variables were reported as median with interquartile range. Bland-Altman was used to compare directly observed and sensor-estimated measurements, as well as in clinic and home measurements. Intraclass correlation coefficients (ICCs) were calculated to assess agreement between directly observed and sensor-estimated measures in the clinic and at home. Linear mixed-effects models for repeated measures (MMRM) compared directly observed and sensor-estimated 6MWD and Cardiac Effort at baseline and Week 8. The MMRM included fixed effects for time, measurement type (clinic or home), and their interaction, with random intercept being individual participants. Changes between in-clinic and home assessment were compared using Spearman correlation coefficient. Analyses were performed in GraphPad Prism 10.3 and R studio 4.5.

2 | Results

2.1 | Baseline Demographics

Twenty-nine participants with pulmonary hypertension completed baseline clinic and home testing. The majority were idiopathic PAH, middle-aged females, and WHO functional class II. Baseline clinic 6MWD and CE were 416 m (300, 472) and 1.75 beats/m (1.48, 2.20), respectively (Table 1). The median home walking space length was 30 ft (30, 35). No safety issues occurred throughout the study, including no device-related complications, falls, tripping on oxygen tubing, syncope, or unexpected emergency room visits related to home 6MWT. Twenty-six participants completed paired week-8 testing; one

TABLE 1 | Baseline demographics.

	$N = 29$
Age, years	57 (44, 68)
Female, %	21 (72)
Body mass index, kg/m^2	32 (25, 41)
NTproBNP, pg/mL	160 (75, 312)
Functional class, I/II/III	1/25/3
Hemoglobin	14 (13, 15)
emPHasis-10	25 (20, 36)
Pulmonary hypertension	
Pulmonary arterial hypertension	
Idiopathic	20
Connective tissue disease	5
Drug and toxin	1
Chronic thromboembolic pulmonary hypertension	3
Therapies	
PDE5I/sGC +/− ERA	29
Inhaled treprostinil	2
Oral or parenteral treprostinil	3/3

Abbreviations: ERA, endothelin receptor antagonist; PDE5I, phosphodiesterase 5 inhibitor; SGC, soluble guanylate cyclase.

participant experienced clinical worsening requiring hospitalization, and two were unable to complete follow-up testing due to weather and COVID-related constraints.

2.2 | Directly Observed and Sensor Estimated 6MWD

Directly observed and lap-estimated 6MWD at baseline and Week 8 are shown in Table 2. Clinic 6MWD values were nearly identical whether measured directly or estimated from lap counts and remained stable over the 8-week interval (Table 3). Agreement between directly observed and lap-estimated 6MWD in clinic was excellent (ICC 0.995; 95% Confidence Interval (CI) 0.990–0.998; $p < 0.0001$). Bland–Altman analysis demonstrated a bias of 0.04% (limits of agreement −9% to 9%) at baseline (Figure 1).

A linear mixed-effects model showed that home 6MWD tended to be approximately 40 m lower than supervised clinic testing over time (Table 4). Agreement between clinic and home lap-estimated 6MWD was moderate to high (ICC 0.909; 95% CI 0.30–0.97; $p = 0.005$). Bland–Altman analysis showed home 6MWD was 13% lower (limits of agreement −37.5% to +10.9%) compared with clinic testing (Figure 1).

2.3 | Baseline Cardiac Effort

One participant had home electrogram data loss at baseline due to device issues. Directly observed, lap-estimated cardiac effort, and heart rate changes at baseline and Week 8 are presented in Table 2. Similar to 6MWD, clinic cardiac

TABLE 2 | Directly observed and lap estimated 6-min walk distance, cardiac effort, heart rate changes, and Borg score at baseline and Week 8.

	6MWD (m)		Cardiac effort (beats/m)		Heart rate (beats/min)						Borg dyspnea index
	Direct	Lap - estimated	Direct	Lap - estimated	Rest	6-min	Peak	Total beats	Reserve (peak −rest)	Recovery (6 min–7 min)	
Clinic baseline	416 (300, 472)	421 (307, 471)	1.75 (1.48, 2.20)	1.74 (1.47, 2.16)	92 (90, 101)	126 (109, 138)	128 (112, 140)	706 (656, 788)	34 (23, 47)	30 (17, 46)	5 (3,7)
Home baseline		343 (245, 423)		1.96 (1.62, 2.71)	87 (80, 95)	115 (108, 133)	118 (111, 135)	669 (634, 750)	32 (22, 41)	11 (6, 19)	4 (2,7)
Clinic Week 8	432 (248, 470)	435 (274, 484)	1.71 (1.53, 2.77)	1.70 (1.50, 2.51)	90 (81, 101)	123 (117, 142)	124 (117, 146)	703 (679, 828)	36 (23, 47)	15 (8, 25)	5 (2,8)
Home Week 8		353 (227, 427)		1.95 (1.61, 2.96)	91 (82, 101)	116 (105, 132)	119 (110, 136)	684 (617, 764)	32 (19, 46)	8 (4, 14)	5 (3, 8)

TABLE 3 | Comparison of directly observed and lap-estimated 6-min walk distance and cardiac effort at baseline and Week 8 using linear mixed-effects models with repeated measures.

	6 min Walk distance			Cardiac effort		
	Estimate	95% Confidence interval	*p* value	Estimate	95% Confidence interval	*p* value
Intercept	389	331–447	< 0.001	2.12	1.69–2.56	< 0.001
Time	2	−14 to 19	0.80	−0.01	−0.19 to 0.17	0.90
Lap estimated	−7	−43 to 29	0.71	0.07	−0.32 to 0.46	0.70
Time × lap estimated	8	−15 to 31	0.51	−0.07	−0.32 to 0.18	0.60

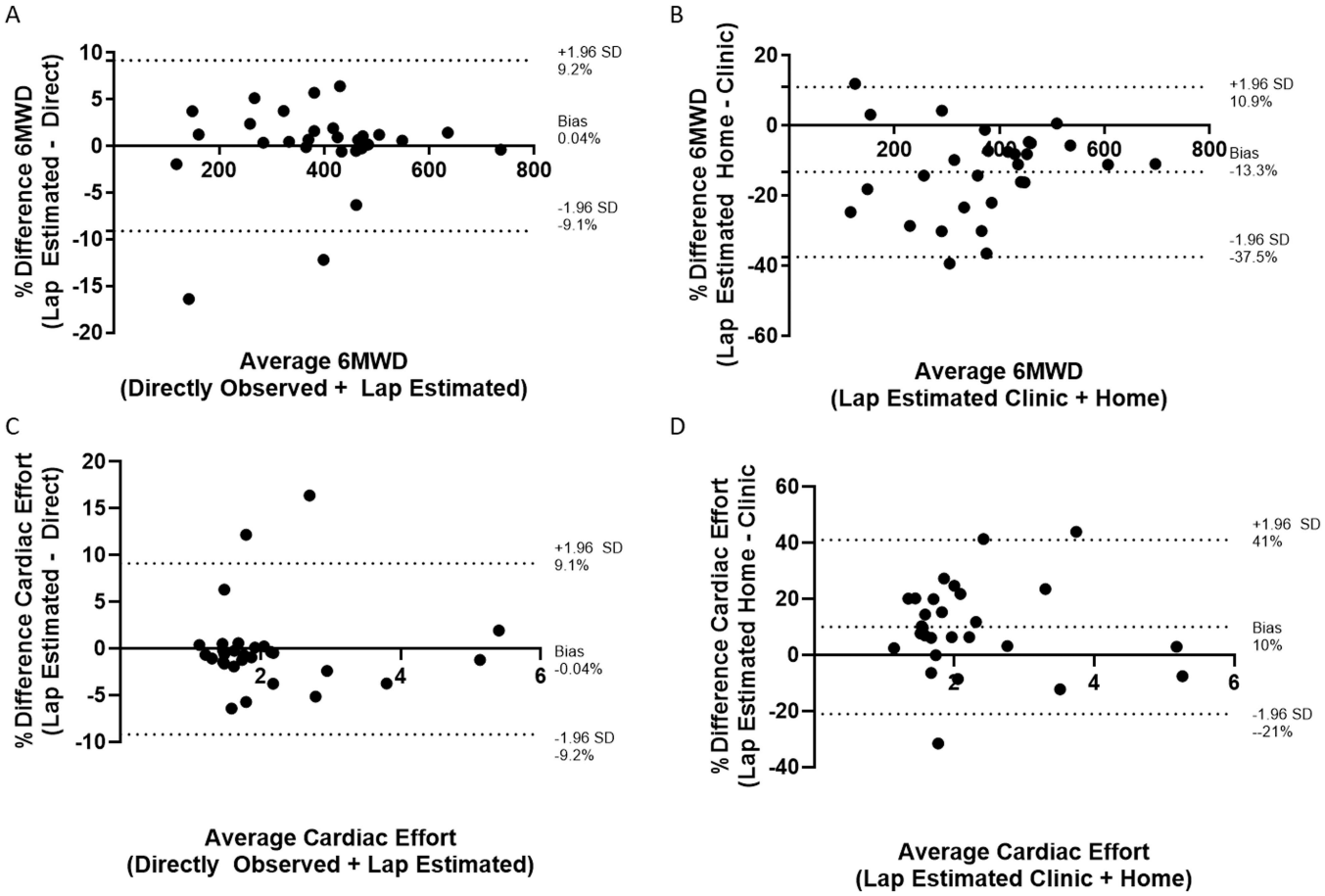

FIGURE 1 | Bland–Altman analyses for baseline lap-estimated 6-min walk distance and cardiac effort. Bland–Altman plot showing agreement between (A) directly observed and lap-estimated 6MWD in the clinic, (B) lap-estimated 6MWD measured in the clinic and at home, (C) directly observed and lap-estimated cardiac effort in the clinic, and (D) lap-estimated cardiac effort measured in the clinic and at home.

effort values measured directly and those estimated by sensors were highly consistent throughout the study (Table 3). Agreement between directly observed and lap-estimated cardiac effort in clinic was excellent (ICC 0.995; 95% CI 0.989–0.997; $p < 0.0001$). Bland–Altman analysis showed a bias of −0.04% (limits −9% to 9%) (Figure 1) at baseline.

Mixed-effects modeling demonstrated no significant difference between home and clinic cardiac effort over time (Tables 2 and 4). Agreement between clinic and home CE was high (ICC 0.904; 95% CI 0.77–0.958; $p < 0.001$). Bland–Altman results indicated that home CE was 10% higher (limits of agreement −21% to +41%) compared to clinic CE at baseline (Figure 1).

2.4 | 8 Week Changes

Using Spearman correlations, changes in directly observed 6MWD were correlated with changes in home 6MWD ($r = 0.40$; 95% CI −0.001–0.69; $p = 0.045$). Changes in directly observed cardiac effort were also correlated with changes in home cardiac effort ($r = 0.42$; 95% CI 0.01–0.71; $p = 0.03$). Linear mixed-effects models demonstrated that changes in

TABLE 4 | Comparison of directly observed and unsupervised home lap-estimated 6-min walk distance and cardiac effort at baseline and Week 8 using linear mixed-effects models.

	6 min Walk distance			Cardiac effort		
	Estimate	**95% Confidence interval**	***p* value**	**Estimate**	**95% Confidence interval**	***p* value**
Intercept	387	329–446	< 0.001	2.09	1.58–2.60	< 0.001
Time	3	−16 to 23	0.74	0.02	−0.22 to 0.25	0.9
Home	−40	−83 to 2	0.06	0.17	−0.34 to 0.69	0.5
Time × home	−5.73	−33 to 22	0.68	0.04	−0.29 to 0.37	0.8

both 6MWD and Cardiac Effort remained directionally consistent over time in clinic and home (Table 4).

3 | Discussion

In this heterogeneous cohort of pulmonary hypertension patients, we found that unsupervised home-based 6MWT performed on a modified walking space was safe and that sensor-estimated home 6MWD and Cardiac Effort approximated in-clinic measurements and tracked in-clinic changes over time. Home 6MWD was 40 m lower than supervised in-clinic 6MWD, likely reflecting the combined impact of additional turns on a shorter walking space, reduced motivation (lower peak heart rate at home), fatigue related to performing multiple 6MWT in a short period of time, slower pacing when walking independently, or a combination of the above. Cardiac Effort helped account for these differences by integrating both distance and physiologic effort. Wearable sensors can quantify functional capacity (maximal effort testing in some participants) with heart rate changes in the home setting that could allow for more frequent assessments between visits. This approach may improve clinical care for patients who travel long distances to specialty centers, facilitate drug titration, and enhance longitudinal risk monitoring.

Including this study, five reports now demonstrate that unsupervised home 6MWTs in pulmonary hypertension can be performed safely, with no events related to tripping on oxygen tubing, syncope, or other complications [7, 12, 13, 20]. As technology has advanced, several approaches have emerged to estimate 6MWD outside of the clinic, including wearables [7, 21], smartphone-based applications [22, 23], and global position satellite [24]. A smartphone app developed for pulmonary hypertension showed an average difference of 15 m (SD 76) compared with directly observed 6MWD [20]. Comparing outdoor 6MWT to indoor clinic-based 6MWD found a mean difference of 2 m (SD 47) [20]. However, weather, baseline mobility, patient preference, and geographic limitations restrict the year-round feasibility of outdoor testing. Chest-based wearable sensors offer a practical indoor alternative. In the present study, we found a negligible bias of 0.04% (limits of agreement −9% to 9%) when comparing lap-estimated to directly observed 6MWD. Estimating distance indoors based on turn counts is reliable as long as the walking-space length is known, and it provides a safe option for higher-risk patients with shorter 6MWD who may have trouble walking outside for 6 min. We found that the difference between home and clinic 6MWD performed within 48 h exceeded the reported minimal clinically important difference (MCID) [25], which can complicate interpretation in isolation. This suggests that home-based reference values on a modified walking space may need to be established, and that the greatest utility of home testing may lie in tracking longitudinal trends within individuals rather than relying on absolute distance thresholds, especially if it is performed in close proximity to a supervised assessment when a clinical change hasn't occurred. Participants also reported that performing the 6MWT in the home was convenient, not intimidating, and the 6MWT task increased their daily movement. In a mixed pulmonary hypertension cohort using a smartphone app, changes in app-estimated outdoors and directly observed 6MWD tracked similarly over time [13], suggesting that digital approaches, whether indoor or outdoor, can capture longitudinal changes in functional capacity.

By incorporating continuous heart rate monitoring into the 6MWT and calculating Cardiac Effort (beats/m), dynamic physiologic changes required to achieve a given 6MWD can be more precisely characterized. Cardiac Effort has been shown to correlate with stroke volume in PAH [5, 6] and demonstrates less variability than 6MWD in stable PAH patients and controls [5]. In the unsupervised home setting, Cardiac Efforts provides context for interpreting changes in estimated 6MWD by distinguishing whether differences reflect altered physiologic reserve (higher heart rate for the same or lower 6MWD) or simply variation in effort (lower heart rate with a decrease in 6MWD). In this study, peak heart rate, heart rate at 6 min, and total heartbeats expended were all lower during both home-based 6MWT compared with supervised 6MWT, suggesting some component of reduced effort likely contributing to the observed 40-m decrease in 6MWD, which cardiac effort helps correct for. Consistent with our prior single-visit observation, cardiac effort remained similar between supervised clinic and unsupervised home settings [14]. In this study, the greatest value of cardiac effort was that changes in cardiac effort observed in the clinic correlated with changes measured at home (Figure 2), and repeated measures showed similar supervised and home measurements (Table 4). This suggests that cardiac effort may serve as a reliable physiologic anchor for home-based testing and offers an opportunity to more closely monitor therapeutic titration (decrease in cardiac effort) or worsening in higher-risk patients (increase in cardiac effort) using wearable sensors.

There is growing interest in incorporating passive physiologic monitoring using wearable devices in pulmonary hypertension and heart failure. Actigraphy has already been included as an endpoint in randomized controlled trials in [26–28], interstitial lung disease [29–31], and heart failure [32]. Across these studies, a consistent finding is low daily activity time, likely driven

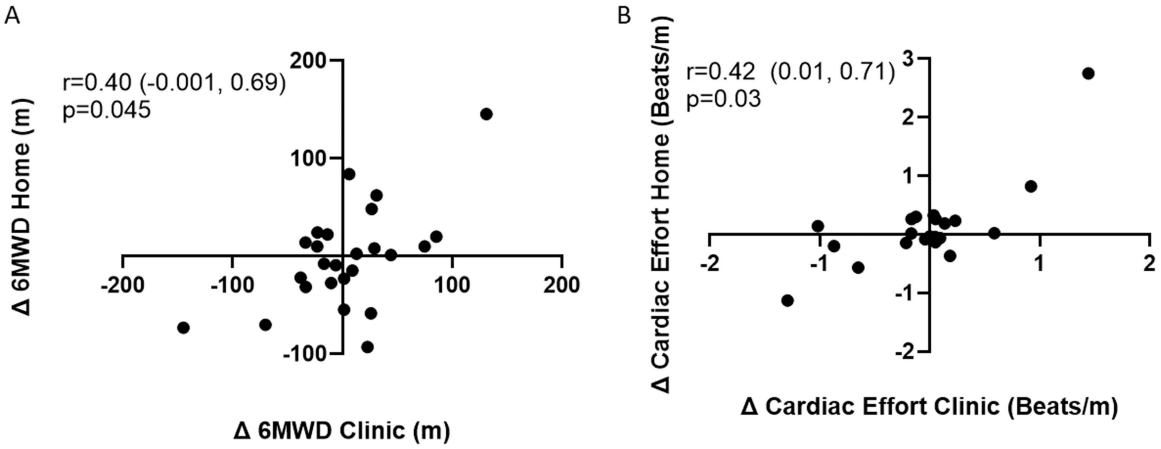

FIGURE 2 | Spearman correlation between changes in supervised 6MWD and cardiac effort with changes in home 6MWD (A) and cardiac effort (B).

by behavioral adaptations, deconditioning, and underlying cardiopulmonary limitations. This low activity volume makes detecting treatment-related changes challenging. In PAH, top bursts of activity, measured as peak 5-min step counts, have increased following initiation of therapy in two uncontrolled studies [33, 34]. More recently, integrating heart rate into daily step analyses has enabled more granular insights into the physiologic demands required to achieve those steps [35]. Metrics such as peak 5-min steps and daily heart rate per step can be conceptualized as passive analogues of an unstructured home 6MWT and cardiac effort. However, passive metrics derived from wrist-based devices are limited by step-classification bias and inaccuracies in photoplethysmography-based heart rate estimation [4, 5, 7]. In contrast, a structured home 6MWT using a chest-based sensor is easier to standardize, interpret, and trend over time. Structured home-based 6MWT may therefore complement passive monitoring by providing a reproducible, physiologically interpretable assessment of functional capacity.

There are limitations to this study. This was a single-center study. Participants were recruited as they presented to the clinic, without a run-in or stability screening period, and therefore changes in clinical status could have occurred. The study was conducted during the COVID-19 pandemic, and pandemic-related scheduling constraints prevented follow-up testing outside of the Week-8 window. Home walking spaces varied across participants and we could not verify the accuracy of course distance or if participants turned early. There could be a learning effect for home walks. Performing three 6MWT in 3 days may be physiologically taxing for some patients and contribute to a lower home 6MWD. Additionally, this study was not designed or powered to evaluate therapeutic response.

In conclusion, this study confirms that wearable devices incorporating electrocardiography and accelerometry enable the 6MWT and Cardiac Effort to be performed safely and reliably in the home setting for patients with pulmonary hypertension. Home-based measurements closely tracked with in-clinic assessments over time, supporting their potential utility for longitudinal monitoring. More frequent testing between clinic visits may allow earlier detection of physiologic changes, provide additional context for symptoms, and help guide therapeutic adjustments. The optimal frequency of home 6MWT to detect clinical changes is unknown. Future studies need to evaluate the balance of patient adherence and analytics to detect meaningful changes. Integration of structured home-based testing with passive monitoring metrics may offer an even more comprehensive understanding of functional capacity. Future studies should evaluate how home 6MWT and Cardiac Effort enhance clinical decision-making, risk assessment, and assess therapeutic response with titratable medications.

Author Contributions

Daniel Lachant: Substantial contributions to the conception or design of the work, analysis, and interpretation of data for the work; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published; agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. **Nishit Agarwal:** Substantial contributions to the analysis and interpretation of data for the work; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published. **Kyle Norton:** Substantial contributions to the data collection; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published. **Cameron McCarthy:** Substantial contributions to the data collection; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published. **Deborah Haight:** Substantial contributions to the data collection; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published. **Sebastian Alphonse:** Substantial contributions to the analysis and interpretation of data for the work; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published. **Melissa Ceruolo:** Substantial contributions to the analysis and interpretation of data for the work; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published. **R. James White:** Substantial contributions to the analysis and interpretation of data for the work; drafting the work or reviewing it critically for important intellectual content; final approval of the version to be published.

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Ethics Statement

Institutional review board approval provided by the University of Rochester Study 0000745.

Conflicts of Interest

Medidata funded the study, and the analysis was performed by URMC. D.J.L., R.J.W., D.H., and K.N. declare no conflicts of interest with Medidata.

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